

check_sparse_tensor_invariants#

https://pytorch.org/docs/stable/generated/torch.sparse.check_sparse_tensor_invariants.html

Description:

A tool to control checking sparse tensor invariants. The following options exists to manage sparse tensor invariants checking in sparse tensor construction:

- Using a context manager:
- Using a procedural approach:
- Using function decoration:

Function Signature

```
class
torch.sparse.check_sparse_tensor_invariants(enable=True)
[source]#

static disable()[source]#

static enable()[source]#

static is_enabled()[source]#
```

Code Examples

```
with torch.sparse.check_sparse_tensor_invariants():  
    run_my_model()
```

```
prev_checks_enabled =  
torch.sparse.check_sparse_tensor_invariants.is_enabled()  
torch.sparse.check_sparse_tensor_invariants.enable()  
  
run_my_model()  
  
if not prev_checks_enabled:  
    torch.sparse.check_sparse_tensor_invariants.disable()
```

```
@torch.sparse.check_sparse_tensor_invariants()  
def run_my_model():  
    ...  
  
run_my_model()
```

```
>>> torch.sparse_csr_tensor([0, 1, 3], [0, 1], [1, 2],  
check_invariants=True)  
Traceback (most recent call last):  
  File "<stdin>", line 1, in <module>  
RuntimeError: `crow_indices[..., -1] == nnz` is not satisfied.
```

Notes & Warnings

NOTE

Note By default, the sparse tensor invariants checks are disabled. Use `torch.sparse.check_sparse_tensor_invariants.is_enabled()` to retrieve the current state of sparse tensor invariants checking.

NOTE

Note The sparse tensor invariants check flag is effective to all sparse tensor constructors, both in Python and ATen.

NOTE

Note Use `torch.sparse.check_sparse_tensor_invariants.enable()` or `torch.sparse.check_sparse_tensor_invariants.disable()` to manage the state of the sparse tensor invariants checks.

Detailed Documentation

Documentation

A tool to control checking sparse tensor invariants.

The following options exists to manage sparse tensor invariants checking in sparse tensor construction:

Using a context manager:

Using a procedural approach:

Using function decoration:

Using `check_invariants` keyword argument in sparse tensor constructor call. For example:

Disable sparse tensor invariants checking in sparse tensor constructors.

See `torch.sparse.check_sparse_tensor_invariants.enable()` for more information.

Enable sparse tensor invariants checking in sparse tensor constructors.

By default, the sparse tensor invariants checks are disabled. Use `torch.sparse.check_sparse_tensor_invariants.is_enabled()` to retrieve the current state of sparse tensor invariants checking.

The sparse tensor invariants check flag is effective to all sparse tensor constructors, both in Python and ATen.

The flag can be locally overridden by the `check_invariants` optional argument of the sparse tensor constructor functions.

Return True if the sparse tensor invariants checking is enabled.

Use `torch.sparse.check_sparse_tensor_invariants.enable()` or `torch.sparse.check_sparse_tensor_invariants.disable()` to manage the state of the sparse tensor invariants checks.